

Small Videogame Development Office Topology

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Network requirements:

-Total employees = 45

-Departments = Production(7), Game Design(9), Art(10), Engineering(16), Audio(3).

-Software in use =

Game Engine:

-Unity

Code:

-Visual Studio Code

Organization, management and Control of versions:

-Git & Github

-Discord

Audio:

-Reaper

-FMOD Studio

3D:

-Blender

-Hardware and equipment needed =

Servers:

-Build / CI Server

-Backup

-Git server

-DNS/DHCP

Hardware:

- 40 High performance pc's
- 7 economic pc's
- 45 or more employee's cellphones/ipads
- (1-45) employees laptops

Building = Single floor office

Logical Design:

VLANs=

- VLAN 10: Engineering (16 devices minimum)
- VLAN 20: Art (10 devices minimum)
- VLAN 30: Game Design (9 devices minimum)
- VLAN 40: Production (7 devices minimum)
- VLAN 50: Audio (3 devices minimum)
- VLAN 60: Servers (4 devices minimum)
- VLAN 70: Employees Wi-Fi (90 devices minimum but we'll do 126 to prevent any extra devices in the long-term)
- VLAN 80: Guests Wi-Fi (between 1-25 devices)

SUBNETS =

Minimum devices needed total count = 119

Rounding off in case the company expands = 512 ipv4 ip's needed

We need a /23 segment for all departments:

192.168.0.0 /23

Applying VLSM:

using 192.168.0.0/23 =

Employees Wi-Fi:

Network Ip: 192.168.0.0 /25

Broadcast ip: 192.168.0.127 /25

Guests Wi-Fi:

Network Ip: 192.168.0.128 /27

Broadcast ip: 192.168.0.159 /27

Engineering:

Network Ip: 192.168.0.160 /27

Broadcast ip: 192.168.0.191 /27

Art:

Network Ip: 192.168.0.192 /27

Broadcast ip: 192.168.0.223 /27

Game Design:

Network Ip: 192.168.0.224 /27

Broadcast ip: 192.168.0.255 / 27

Production:

Network Ip: 192.168.1.0 /28

Broadcast ip: 192.168.1.15 /28

Servers:

Network Ip: 192.168.1.16 /28

Broadcast ip: 192.168.1.31 /28

Audio Engineers:

Network Ip: 192.168.1.32 /29

Broadcast ip: 192.168.1.39 /29